

SentinelWire™ IOC Extractor v1.0 (PowerShell + tshark)

Explanation & Usage Guide

What this script is

extract_iocs_UPDATED_v1.1.ps1 is a defensive SOC utility that extracts Indicators of Compromise (IOCs) from a **PCAP/PCAPNG** file using **tshark** (Wireshark CLI). It is designed to complement SentinelWire Wireshark profiles by turning packet captures into a **structured IOC output set** plus **automatic prioritization**.

Unlike basic IOC dump scripts, SentinelWire v1.1 adds two SOC-grade layers:

1. **IOC Confidence Scoring (0–100)** — ranks IOCs by likely relevance
2. **MITRE ATT&CK Mapping** — attaches behavior-based technique tags to each IOC

This helps analysts go from “raw artifacts” to “triage-ready intelligence” quickly.

Key features (v1.1)

-  Creates a unique **run folder** for each execution:
C:\IOCs\run-YYYYMMDD-HHMMSS\
-  Extracts:
 - IPv4/IPv6 IPs
 - DNS queries
 - HTTP hosts and User-Agents (if cleartext HTTP exists)
 - TLS SNI (Server Name Indication)
 - JA3/JA3S fingerprints (if available in your Wireshark build)
 - Base64-like payload candidates (triage aid)
-  Produces **scored outputs**:
 - ioc_scored.csv
 - ioc_scored_top20.txt
 - ioc_scored_summary.txt

Requirements

1) Windows

- Windows 10 / 11

2) Wireshark + tshark

- Install Wireshark and ensure **tshark.exe** is available in PATH.

Test in PowerShell:

```
tshark -v
```

If this returns a version string, you're good.

3) PowerShell Execution Policy

Set once (CurrentUser scope is sufficient):

```
Set-ExecutionPolicy -Scope CurrentUser -ExecutionPolicy RemoteSigned
```

Recommended folder structure

- Script location:
C:\scripts\
- Output root:
C:\IOCs\
- Captures folder:
C:\captures\

How to run (step-by-step)

1. Open **Windows PowerShell**
2. Go to the scripts directory:

```
cd "C:\scripts"
```

3. Run the extractor against a PCAP/PCAPNG:

```
.\extract_iocs_UPDATED_v1.1.ps1 "C:\captures\case_20251209.pcapng"
```

Where outputs are saved

Each run creates a folder like:

```
C:\IOCs\
```

```
run-20251216-093012\
```

```
ipv4_ips.txt
```

```
ipv6_ips.txt
```

```
dns_queries.txt
```

```
http_hosts.txt
```

```
http_user_agents.txt
```

```
tls_sni.txt
```

ja3_fingerprints.txt

base64_candidates.txt

ioc_scored.csv

ioc_scored_top20.txt

ioc_scored_summary.txt

How scoring works (IOC Confidence Score: 0–100)

The score is an **analyst prioritization hint**, not a verdict.

Scoring factors:

- **Persistence** (repeated appearance) → increases score
- **Correlation** (appears in multiple categories) → increases score
- **IOC type risk** (e.g., long DNS domains) → increases score
- **Known-good reduction** (Google/Microsoft/CDN patterns) → reduces score

General interpretation guideline:

- **0–29** → Low confidence (log/monitor)
- **30–69** → Medium confidence (enrich + correlate)
- **70–100** → High confidence (triage priority + escalation candidate)

You can customize the **KnownGoodPatterns** list inside the script for your environment.

How MITRE ATT&CK mapping works

MITRE mapping is **behavior-based classification**, not attribution.

Examples:

- Long DNS or DNS anomalies → **T1071.004 (DNS C2)**
- MITM/ARP behaviors → **T1557.002 (ARP Cache Poisoning)**, **T1040 (Network Sniffing)**
- SYN anomalies/scanning → **T1046 (Network Service Scanning)**
- TLS anomalies → **T1573 (Encrypted Channel)**, **T1071.001 (Web C2)**

MITRE technique tags appear in:

- ioc_scored.csv (MITRE column)
- ioc_scored_top20.txt

- ioc_scored_summary.txt

What to review first (SOC triage order)

If you want “fastest value first”, review:

1. ioc_scored_top20.txt  (fastest triage)
2. ioc_scored_summary.txt  (case overview)
3. ioc_scored.csv  (full dataset)

Then pivot into raw artifacts as needed:

- dns_queries.txt for suspicious domains
- tls_sni.txt for suspicious encrypted destinations
- http_user_agents.txt for unusual clients
- ja3_fingerprints.txt for fingerprinting (when available)

Common issues & fixes

“PCAP file not found”

- The file path is wrong. Verify:

```
dir "C:\captures"
```

“tshark is not available in PATH”

- Wireshark not installed or PATH not set.
- Reinstall Wireshark or add tshark.exe to PATH, then test:

```
tshark -v
```

JA3 output is empty

- Some Wireshark/tshark builds do not expose JA3 fields by default. This is normal. The script will still work.

Base64 candidates look noisy

- This is expected. The base64 scan is a **triage signal**, not a detection verdict. Always correlate with destinations and timing.

Legal / operational note

This tool is intended for **authorized defensive incident response only**.

Do not capture or analyze traffic from networks you do not own or where you do not have explicit permission.